

PS70 CycloneWatch – Team Task Sheet

8 Days · Day 4 = Mentor Checkpoint (Proof of Concept) · Day 8 = Final Demo

Shared Context (read once, everyone)

What we're building: An AI/ML system that identifies, classifies, and predicts tropical cyclone patterns from multi-source satellite data (IR, visible, water vapour) – no NWP model inputs, satellite-only by design.

Positioning line (memorize this, everyone will get asked):

"We are not replacing IMD. IMD uses complex NWP physics models. We are building a decision-support layer that automates the interpretation of raw satellite data to classify structural patterns and predict evolution – addressing MoES's acknowledged gaps in short-fuse event interpretation and uncertainty visualization."

Timeline:

- Days 1–4 → working proof-of-concept, shown to mentor
- Days 5–8 → polish into judge-ready version, using mentor feedback from Day 4

Golden rule: if you're ever blocked waiting on someone else's output, say so immediately – don't sit on it. Fake/stub data is fine for Day 1–2 as long as it matches the agreed shape.



ML – Aditya Tyagi (ECE)

You're carrying two models solo – classification then prediction, in that order, never both at once. Give Backend and Frontend a stub output on Day 1 so they're never blocked on you.

- **Day 1:** Scaffold PyTorch environment. Build dataloaders that read the multi-band files Data is producing. Give Backend a fake/stub classification output (fixed JSON) so they can build against it immediately.
- **Day 2:** Train/deploy the CNN → outputs (1) center lat/long, (2) pattern category, (3) confidence score. Hand the real model to Backend to wire in.

- **Day 3:** Deploy the temporal model → ingest T-12→T0 satellite frames, output predicted center + pattern state for T+12/T+24. Hand off to Backend.
 - **Day 4:** No new modeling. Bug fixes only. Write an honest one-paragraph "known limitations" note for documentation.
 - **Day 5:** Improve the classification model first (more data/epochs/augmentation) based on mentor feedback.
 - **Day 6:** Improve the temporal model. Calibrate confidence outputs so the uncertainty cone reflects real model confidence, not a placeholder.
 - **Day 7:** Freeze the models – no more training changes after this point.
 - **Day 8:** Full rehearsal. Be ready to explain the model choices and limitations in Q&A.
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Backend – Satyam Srivastava (IT) – Team Leader

In addition to your Backend tasks below, you're also the point of contact for daily status updates, unblocking people, and keeping the whole team pointed at the Day 4 and Day 8 targets.

You're the glue between ML, Frontend, and App – lock the API contract early and don't change its shape without telling everyone.

- **Day 1:** Scaffold FastAPI + PostgreSQL/PostGIS in Docker Compose. Expose a basic endpoint serving satellite image tiles. Agree on the API contract (request/response shapes) with Frontend + App Dev.
 - **Day 2:** Build `/api/ps70/classify` using ML's real model output. Store classifications in PostGIS as a retrievable time-series.
 - **Day 3:** Build the forecasting endpoints for ML's temporal model. Generate an uncertainty-cone polygon from the model's confidence intervals.
 - **Day 4:** Finalize `/api/replay` – must serve data as if the system were running live in the past, using pre-loaded historical demo cases so nothing depends on live internet during the demo.
 - **Day 5:** Fix any backend/API issues the mentor flagged. Start the real uncertainty-cone calculation if not already solid.
 - **Day 6:** Finalize uncertainty-cone geometry. Add a dedicated `/api/metrics` endpoint (accuracy/MAE numbers).
 - **Day 7:** Freeze the API. Stress-test the replay endpoint on all demo cases – zero live-internet dependency.
 - **Day 8:** Full rehearsal, live with Frontend and App Dev.
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App Dev – Aniket Tiwari (ECE)

Deliberately smaller scope than the web dashboard – a status view and an alert list, not a full replica.

- **Day 1:** Project skeleton (Flutter/React Native). Confirm the API contract with Backend. Build static screen shells.
 - **Day 2:** Connect to the real `/classify` endpoint. Show live classification/status on the app.
 - **Day 3:** Add predicted-position/alert display. Build basic offline fallback with one bundled demo case (don't rely on venue wifi).
 - **Day 4:** Freeze features. Fix bugs only. Confirm the app runs reliably on the actual device you'll demo on.
 - **Day 5:** Address any app-specific feedback from the mentor.
 - **Day 6:** Visual polish pass — match the web dashboard's now-finalized visual identity (coordinate with Designer+Frontend).
 - **Day 7:** Freeze app features.
 - **Day 8:** Full rehearsal on the real device.
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Designer + Frontend — Kavya Agarwal (IT)

You're doing two jobs in one — the biggest schedule risk on the team. Don't over-spend Day 1 on design; lock the visual identity fast and get the shell running. Your real polish pass is scheduled for Day 6, not Day 1-4 — plan your energy accordingly.

- **Day 1:** Lock design tokens/visual identity quickly. Build the React + Leaflet shell with layer toggles (Infrared / Visible / Water Vapor).
 - **Day 2:** Render classification output on the map — pulsing center marker + Level 2 data card (pattern type + confidence).
 - **Day 3:** Visualize the predicted future track on the map. Build a timeline UI showing predicted pattern evolution.
 - **Day 4:** Build the Historical Replay slider (T-48h → T0, predicted vs. actual). Build the Level 3 evidence panel (click to see the raw satellite band behind each decision). It's OK if this looks rough for the mentor checkpoint — full polish comes Day 6.
 - **Day 5:** Address UI confusion the mentor flagged.
 - **Day 6:** Full visual polish pass — protect this day, this is where the "real" design work happens. Finish the Level 3 evidence panel properly.
 - **Day 7:** Freeze features. Only visual/bug fixes from here.
 - **Day 8:** Full rehearsal, live with Backend.
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Data — Abhinav Pal (IT)

The technically heavier of the two "flex" roles — you're wrangling raw multi-band satellite files, not simple spreadsheets.

- **Day 1:** Download raw multi-source satellite sequences (IR, water vapour) for the 2 events Research locks in. Standardize formats (NetCDF/GeoTIFF) so ML can load them directly. Satellite-only — no NWP data.
 - **Day 2:** Keep standardizing additional event data ahead of ML's Day 3 prediction work so ML is never blocked.
 - **Day 3:** Prep historical ground-truth imagery needed for Day 4 validation.
 - **Day 4:** Work with Research to calculate the exact Mean Absolute Error of predicted center vs. IMD Best Track data. Use the real number, don't round up.
 - **Day 5:** Expand the historical event set beyond the 2 demo cases. Fix any data-quality issues the mentor flagged.
 - **Day 6:** Support ML's confidence-calibration work with any additional data needed.
 - **Day 7:** Run the full backtest across the expanded event set with Research; compute final MAE/accuracy figures.
 - **Day 8:** Be ready to explain exactly where the data came from if asked.
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Research — Arshit Singh (CS)

The lighter, structured-data "flex" role — best-track CSVs and domain knowledge, not raw imagery pipelines.

- **Day 1:** Lock in 2 historical cyclone events (e.g. Biparjoy). Scrape/pull IBTrACS best-track CSVs for those events. Draft the pattern-classification taxonomy (eye / banding / shear-affected / etc.) for ML.
 - **Day 2:** Finalize ground-truth labels for the locked events. Start the MoES/IMD gap-analysis writeup.
 - **Day 3:** Prep the historical ground-truth position data Data will need for Day 4 validation.
 - **Day 4:** Work with Data to calculate the MAE of predicted vs. actual center positions. Support documentation drafting.
 - **Day 5:** Expand the historical event set (more storms = more convincing accuracy claim) alongside Data.
 - **Day 6:** Draft the "known limitations" and "what MoES/IMD gap this addresses" slide content using real numbers.
 - **Day 7:** Help finalize the backtest and the written documentation + PPT content.
 - **Day 8:** Own the Q&A defense sheet — be ready to field "why not just use IMD," "where's your data from," "can this scale."
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Everyone, All 8 Days

- End each day with a one-line status update to the group (Satyam is tracking overall progress): what's done, what's blocked.
- Nothing gets cut from scope without telling Research so the "next steps" slide stays accurate.
- Day 4 checkpoint: bring the honest state — what works, real numbers, known limitations. That's what a checkpoint is for.
- Day 8: everyone should be able to explain the *whole* system, not just their own piece, in case a judge/mentor asks the wrong person.